

Programming Using Numbers (part 1)

CMSC 104 Section 2
September 9, 2026

Administrative Notes

Classwork 2 is due tonight, before midnight

Project 1 is assigned today

- It is due in two weeks - Wednesday, September 23
- We'll talk about it at the end of class. You don't yet have all the material you need to do it. Next Monday's lecture will cover some more needed material.
 - Go ahead and work on it, especially Part 2 of the assignment.

Quiz 1 is ***Monday, September 21***

- I hate giving exams on Mondays. I try to have a review session on Monday, and the quiz on Wednesday, but with Labor Day there was no way to make the dates work this semester.
- There will be a "sample quiz" posted on the class GitHub repo, by no later than next Monday, the 14th.
- Class on the 16th will include a review session - going over the sample quiz, and answering any other questions you may have
 - The material isn't that difficult, but a lot of students don't have experience with the kind of "open book" exams I give. So we'll work through that.
- If you qualify for accommodations and want to take the quiz at SDS, please make arrangements with them and I will provide them the quiz.

The python “Main program” and functions

We’re jumping ahead a little bit, but I need to make sure you’re aware of the terms “main program” and “function”

When a Python program starts executing: the interpreter looks for the start of the main program

- The first instruction in the main program is the first instruction that gets executed by the computer
- Functions only get executed when you call them - later in the semester

If there’s no explicit marking of “main program” and “function” the Python interpreter assumes the whole program is the main program

A digression - main programs and functions

The textbook is obsolete `def main()` went out a few years ago

- Here is how Python handles the main program currently:

```
if __name__ == "__main__": #there are four pairs of underscores  
    # the rest of the main program should be indented
```

So far we haven't written any functions, but there's no harm getting in the habit of identifying where your main program starts

So we will generally start the executable parts of our programs with the line above

Integers vs real numbers

Python recognizes two different types of numbers;

- Integers - whole numbers; numbers without a decimal place or an exponent
 - Positive, negative or zero
 - Python as a language does recognize a “maximum integer” - it depends on what your computer can store
- Real numbers - have decimal places and/or exponents
 - 6.02e23 or 6.02E23 - in this once instance Python is not case sensitive

Valid operations for Integers

Operator	What it does	example
+	addition	$5 + 6 = 11$
-	subtraction	$13 - 9 = 4$
*	multiplication	$5 * 8 = 40$
**	exponentiation	$2^{**}4 = 16$
//	Integer division - yields only the integer quotient, not the remainder	$4//2 = 2$ $5//2 = 2$
%	Modulus - yields only the remainder of integer division, not the quotient	$4\%2 = 0$ $5\%2 = 1$

Valid operations for floating point (real) numbers

Operator	What it does	example
+	addition	$5.2 + 6.3 = 11.5$
-	subtraction	$13.8 - 9 = 4.8$
*	multiplication	$5.3 * 8 = 42.4$
**	exponentiation	$2.2^{**}4 = 23.4256$
/	Floating point division	$5/2 = 2.5$

Some examples

Calculate $5!$

Calculate $5.0!$

Order of operations

What is the value of the expression:

$$5+9*3**2-7//4$$

The answer is 85. But how did we get that?

The rules Python enforces are similar to what you learned in math:

- Anything in Parentheses comes first. Start with the innermost parentheses, and work out. Go left to right if there are multiple, separate sets of parentheses
-

Order of operations:

After you've handled all parentheses:

- Next comes exponentiation.
 - If you have multiple exponents go **right to left**. That is,

$4^{**}3^{**}2$ is $4^{**}9 = 262144$. It's NOT $64^{**}2 = 4096$

- Then comes multiplication and division.
 - Go **left to right** in the expression
 - Both integer division (//) and modulus (%) count as “division”
- Last comes addition and subtraction
 - Again, go **left to right** in the expression

Casting variables - changing the type of a variable

Automatic conversion by the system - int to floating point

- Let's look at what happens with $5/2$ and $5//2$
- The difference is that with floating point division, Python automatically casts - changes the type of - the result to floating point

Casting using `int()`, `float()`, and `str()`

- You can also force a cast - a change in type - yourself

Syntax of a “for i” loop

```
for i in range(start, stop, step):  
    # do some things  
  
    # whatever is indented under the “for” statement  
  
    # is part of the loop
```

Doing math with input from the keyboard

Remember from last week that all input from the keyboard is read as a string

- You can't do math on strings in Python!!
- So you have to cast the variable you read in to an integer or a floating point, as is needed

```
# ask the user to input a positive integer
#find the sum of all the numbers between 1 and the
number that the user entered
sum = 0
print("Welcome to my program. Enter a positive integer ")
print(" and we will print out the sum off all the positive integers")
print("up to and including your number")
num = input("Please enter a positive integer: ")
for index in range(1,num+1,1):
    sum = sum + index
print("The answer is ", sum)
print("Thanks for playing. Isn't this cool?")
```

```
# ask the user to input a positive integer
#find the product of all the numbers between 1 and the
number that the user entered
product = 1
print("Welcome to my program. Enter a positive integer ")
print(" and we will print out the product of all the positive integers")
print("up to and including your number")
num = int(input("Please enter a positive integer: "))
for index in range(1,num+1,1):
    product = product * index
print("The answer is ", product)
print("Thanks for playing. Isn't this cool?")
```